

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA0910	Course Title:	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	Assessment:	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	CO1: <i>Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)</i>		
Student Name:	B.Priyanka	Reg. No.:	192421206
Section / Batch:	B	Date:	27-07-2026

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.

Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation. Each question carries 20 Marks.

For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.

No calculators or electronic devices permitted during the viva session.

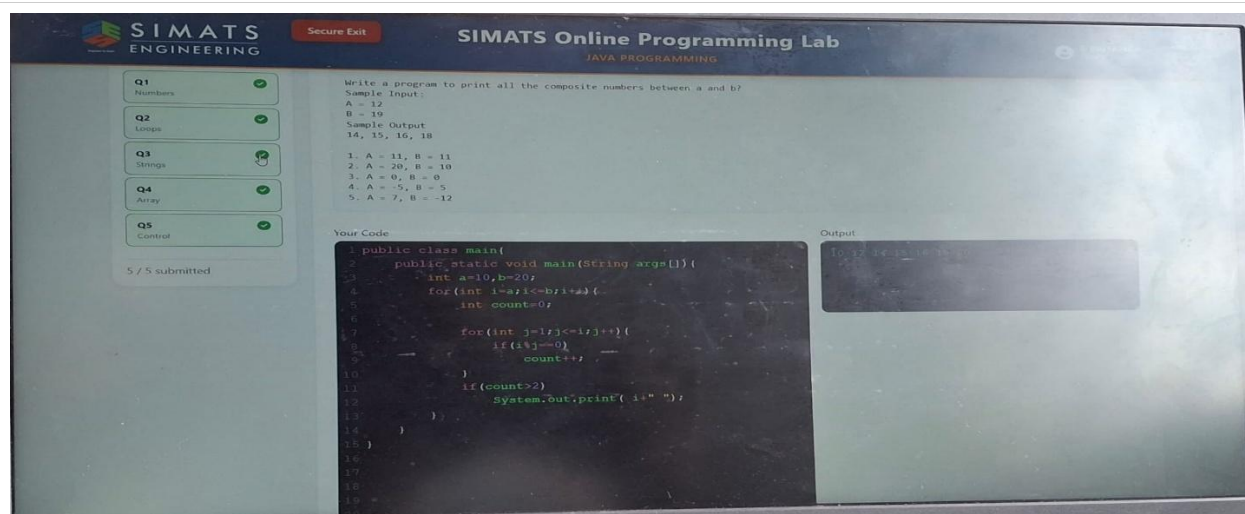
Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
GRAND TOTAL:			100 Marks

SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing) [20 Marks]

```
START
INPUT a, b
FOR i = a TO b
  IF i > 3 THEN
    composite = false
    FOR j = 2 TO  $\sqrt{i}$ 
      IF i % j == 0 THEN
        composite = true
        BREAK
      END IF
    END FOR
    IF composite == true THEN
      PRINT i
    END IF
  END IF
END FOR
STOP
```

Answer / Program with output:



Q2. (Level 1 – Program Writing) [20 Marks]

```
START
INPUT n
FOR i = 1 TO n
  FOR j = 1 TO i
    PRINT "& "
  END FOR
  PRINT NEW LINE
END FOR
STOP
```

Answer / Program with output:

The screenshot shows the SIMATS Online Programming Lab interface. On the left, a sidebar lists questions under 'LEVEL 1 - EASY', with Q1 (Numbers) selected. The main area displays a problem titled 'Level 1 - Easy: (E) RT-Pattern - Loops'. The problem asks to write a program to print a pattern based on sample input: 'Enter the Character to be printed: %', 'Max Number of time printed: 3', and the output pattern: '% % %', '% %', '% % %'. The 'Your Code' section contains a Java program that takes a character and a number as input and prints the corresponding pattern. The 'Output' section shows the result of running the code.

QUESTIONS
LEVEL 1 - EASY
Q1 Numbers ✓
Q2 Loops ✓
Q3 Strings ✓
Q4 Array ✓
Q5 Control ✓
5 / 5 submitted

Level 1 - Easy: (E) RT-Pattern - Loops ✓ Submitted

Write a program to print the following pattern
Sample Input:
Enter the Character to be printed: %
Max Number of time printed: 3
% % %
% %
% % %

Your Code

```
1 public class Main{  
2     public static void main(String args[]){  
3         int n=3;  
4         for(int i=1;i<=n;i++){  
5             for(int j=1;j<=i;j++){  
6                 System.out.print("%");  
7             }  
8             System.out.println();  
9         }  
10    }  
11 }  
12 }
```

Output

```
% % %  
% %  
% % %
```

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Q3. (Level 1 – Program Writing) [20 Marks]

START
INPUT word
FOR i = LENGTH(word) - 1 TO 0
 PRINT word.charAt(i)
END FOR
STOP

Answer / Program with output:

The screenshot shows the SIMATS Online Programming Lab interface. On the left, a sidebar lists questions under 'LEVEL 1 - EASY', with Q3 (Strings) selected. The main area displays a problem titled 'Level 1 - Easy: (E) String Reverse - Strings'. The problem asks to write a program to reverse a word using a loop, with a note not to use inbuilt functions. The sample input is 'String: TEMPLE' and the sample output is 'Reverse String: ELPMET'. The 'Your Code' section contains a Java program that takes a string as input and prints its reverse. The 'Output' section shows the result of running the code.

LEVEL 1 - EASY
Q1 Numbers ✓
Q2 Loops ✓
Q3 Strings ✓
Q4 Array ✓
Q5 Control ✓
5 / 5 submitted

Level 1 - Easy: (E) String Reverse - Strings ✓ Submitted

Write a program to reverse a word using loop? (Not to use inbuilt functions)
Sample Input:
String: TEMPLE
Sample Output:
Reverse String: ELPMET
1. SIGN UP
2. AT LEAST
3. 1245
4. 10055
5. 145*999=144855

Your Code

```
1 public class Main{  
2     public static void main(String args[]){  
3         String str="TEMPLE";  
4         String rev="";  
5         for(int i=str.length()-1;i>=0;i--){  
6             rev=rev+str.charAt(i);  
7         }  
8         System.out.println(rev);  
9     }  
10 }
```

Output

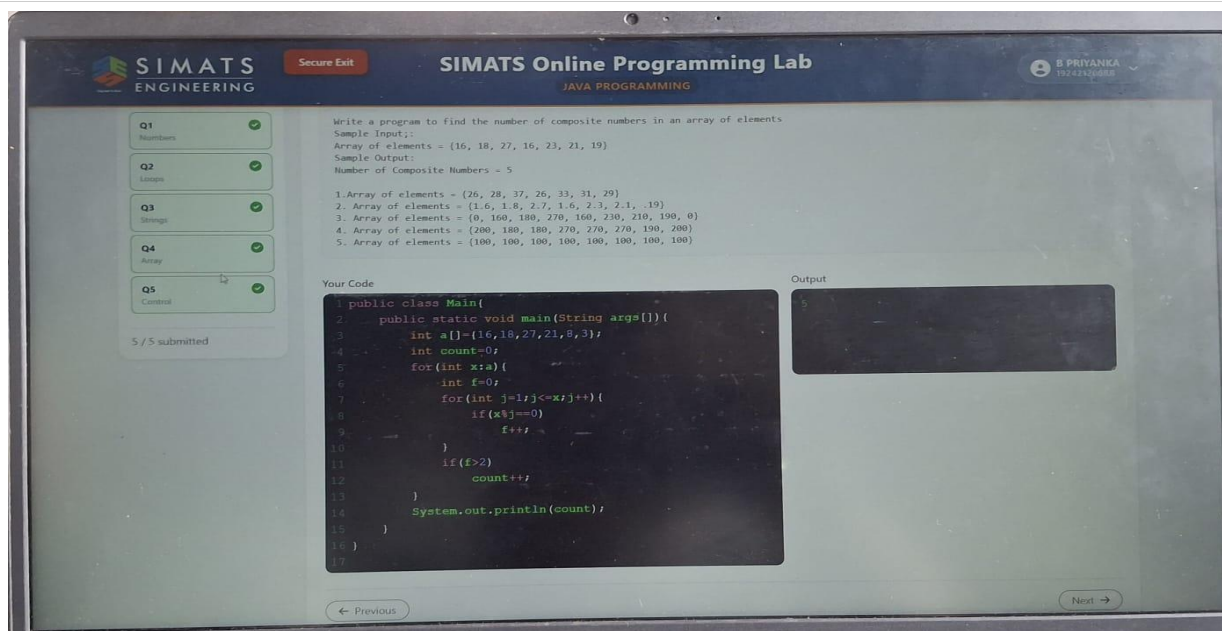
```
ELPMET
```

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Q4. (Level 1 – Program Writing) [20 Marks]

```
START
INPUT n
DECLARE arr[n]
FOR i = 0 TO n-1
    INPUT arr[i]
END FOR
count = 0
FOR i = 0 TO n-1
    IF arr[i] > 1 THEN
        divisor = 0
        FOR j = 1 TO arr[i]
            IF arr[i] % j == 0 THEN
                divisor = divisor + 1
            END IF
        END FOR
        IF divisor > 2 THEN
            count = count + 1
        END IF
    END IF
END FOR
PRINT count
STOP
```

Answer / Program with output:



Q5. (Level 1 – Program Writing) [20 Marks]

```
START
FUNCTION simpleInterest(P, R, T)
    SI = (P * R * T) / 100
    RETURN SI
END FUNCTION
INPUT P, R, T
result = simpleInterest(P, R, T)
PRINT "Simple Interest = ", result
STOP
```

Answer / Program with output:

The screenshot displays the SIMATS Online Programming Lab interface. On the left, a sidebar lists topics: Q1 Numbers, Q2 Loops, Q3 Strings, Q4 Array, and Q5 Control. The main area contains a problem statement for calculating simple interest, sample input/output, and a list of test cases. Below this, the 'Your Code' section shows a Java program with a `simpleInterest` function and a `main` method. The 'Output' section shows the result: 'Simple Interest = 100'.

Problem Statement: Write a program using function to calculate the simple interest. Suppose the customer is a senior citizen. He is being offered 12 percent rate of interest; for all other customers, the ROI is 10 percent.

Sample Input:
Enter the principal amount: 200000
Enter the no of years: 3
Is customer senior citizen (y/n): n
Sample Output:
Interest: 60000

Test Cases:

- Principal: 2000 , Years: 0
- Principal: 20000 , Years: -2
- Principal: -2000 , Years: 2
- Principal: 2 , Years: 2000
- Principal: 0 , Years: 5

Your Code:

```
1 public class Main{
2     static void simpleInterest(int p,int r,int t){
3         int si=(p*r*t)/100;
4         System.out.println("Simple Interest = "+si);
5     }
6     public static void main(String args[]){
7         int p=1000;
8         int r=5;
9         int t=2;
10        simpleInterest(p,r,t);
11    }
12 }
```

Output:
Simple Interest = 100

RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____